



PieCIng an Express

This year, bits and bytes are in for some lane driving...



Sunday, 9th May, 2004: "ATI readies PCI-Express versions of X800 Pro, XT." 13th May, 2004: "SiS has announced that several motherboards based on their SiS965L chipset, which features PCI-Express technology, will be available in June." It's all happening for PCI Express, and 2004 will be the year when it really takes off. But what is PCI-Express anyway?

Think of PCI-Express as an 'evolutionary' upgrade to the PCI bus. The PCI bus has been with us for a long time, since the early 1990s. In terms of I/O, your PC hasn't progressed much since the 486. All the data that is passed by and between your LAN card, hard drive, PCI graphics card, sound card and so on, still uses the same I/O system—the PCI bus. And all traffic between these devices must share the same bandwidth.

Why does this need to change? It's because of emerging technologies, and the need for more flexibility in component design. The patience of the SCSI, RAID and Gigabit Ethernet data transfer technologies is being tested by the PCI bus, which typically operates at 33 MHz, and transfers a maximum of 133 MB of data per second. There are different standards for the PCI bus, but these speeds are typical. And the aforementioned technologies can transfer data at much faster speeds than that, but they aren't being allowed to.

PCI-Express was approved as a standard in April 2002, introduced by Intel in partnership with companies such as Microsoft, IBM, Dell, HP and others. It is fully backward-compatible with the old PCI bus standard. Data on the PCI-Express bus will travel in packets, like it does on Ethernet. The bus will not be a parallel bus connecting everything like a single highway, and through

which all data of all kinds passes at one, fixed rate, which is what happens on the PCI bus. PCI-Express will consist of an assembly of serial—not parallel—lanes, wired from point to point, and clocked individually depending on device. Each lane will consist of two pairs of data lines, and each lane will be capable of 32X data transfer rate in each direction—1X is 2 Gbps. Slots for graphics cards will be modular—16X slots will replace the AGP 8X standard. You'll be able to plug in a 16X card into a 32X slot, because of the modularity of the standard.



What does this mean? Try and imagine it this way: whereas previously your computer was a fixed-speed highway with homes all along the way, it'll now be a town with homes, and roads of different widths between them. To go from one house to another, you wouldn't need to traverse the highway, coping with other traffic—you could take a shortcut. You could choose a less clogged road. You could drive at different speeds, too.

This translates to more flexibility, more bandwidth, more speed—all of which stem from the basic architecture of the standard. Intel has even hinted at a futuristic vision for servers and desktops—one that would separate the computing unit from the monitor, interface devices and external drives using PCI-Express cable connections as a go-between. Think of this as a computer more like a stereo system, with components stacked one on top of the other, with cables going between them.

Intel's Grantsdale chipset supports PCI-Express. So will Longhorn. In fact, graphics cards, chipsets, motherboards, operating systems, you name it—they'll all be jumping on the PCI-Express bandwagon, because it simply will be the PC bus standard for some years to come.



PEOPLE TO WATCH

Indian-American Tapesesh Yadav, Founder and CEO of NanoProducts Corporation, Longmont, Colorado, has 28 patents in nanotechnology, and more than 30 pending. Dr Yadav obtained his PhD from MIT; his thesis in carbon nanotechnologies and related deliberations provided the inspiration for NanoProducts Corporation (NPC). NPC offers a wide range of nanoscale powders and nanoscale powder-derived products, manufactured using patented processes.

"The first generation of nanotechnology will just be performance enhancements to existing micro-materials, but it's the second generation, in about three years, that will start employing nano-materials in much more significant quantities," says Yadav.

Tapesesh Yadav
CEO, NANOPRODUCTS CORPORATION

